

Exercise Sheet #9: AC Power and Resonances

Discussion: Dec 15, 2025

Exercise 1: Instantaneous and Average Power

If $v(t) = 160 \cos(50t)$ V and $i(t) = -33 \sin(50t - 30^\circ)$ A, calculate the instantaneous power and the average power.

Solution

First rewrite the current using

$$\sin x = \cos(x - 90^\circ) :$$

$$i(t) = -33 \cos[(50t - 30^\circ) - 90^\circ] = -33 \cos(50t - 120^\circ).$$

The instantaneous power is

$$p(t) = v(t)i(t) = 160 \cos(50t) [-33 \cos(50t - 120^\circ)] = -5280 \cos(50t) \cos(50t - 120^\circ).$$

Use the identity

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)].$$

Let $A = 50t$, $B = 50t - 120^\circ$:

$$p(t) = -5280 \cdot \frac{1}{2} [\cos(120^\circ) + \cos(100t - 120^\circ)] = -2640 [\cos 120^\circ + \cos(100t - 120^\circ)].$$

Since $\cos 120^\circ = -\frac{1}{2}$,

$$p(t) = -2640 \left[-\frac{1}{2} + \cos(100t - 120^\circ) \right] = 1320 - 2640 \cos(100t - 120^\circ).$$

Thus the **instantaneous power** is

$$p(t) = 1320 + 2640 \cos(100t + 60^\circ) \text{ W}.$$

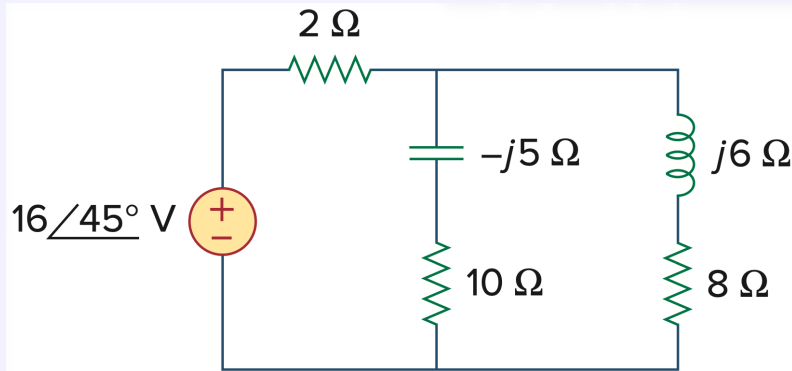
The **average power** is the DC term:

$$\bar{P} = 1320 \text{ W}.$$

Exercise 2: Complex Power Calculation

For the entire circuit below, calculate:

- the complex power
- the average power delivered by the source
- the reactive power
- the apparent power
- the power factor



Solution

Two parallel branches after the 2Ω resistor:

$$\underline{Z}_1 = 10 - j5, \quad \underline{Z}_2 = 8 + j6.$$

Total impedance (including the 2Ω resistor in series):

$$\begin{aligned} \underline{Z} &= 2 + \left(\frac{1}{\underline{Z}_1} + \frac{1}{\underline{Z}_2} \right)^{-1} = 2 + \left(\frac{1}{10 - j5} + \frac{1}{8 + j6} \right)^{-1} \\ &= 2 + \left(\frac{10 + j5}{100 + 25} + \frac{8 - j6}{64 + 36} \right)^{-1} = 2 + \left(0.08 + j0.04 + 0.08 - j0.06 \right)^{-1} \\ &= 2 + \frac{0.16 + j0.02}{0.16^2 + 0.02^2} = 2 + \frac{0.16 + j0.02}{0.026} = \frac{0.212 + j0.02}{0.026} \Omega \\ Z &= \frac{\sqrt{(0.212^2 + 0.02^2)}}{0.026} \approx 8.1901 \Omega \\ \varphi &= \arctan \frac{0.02}{0.212} = 5.39^\circ \end{aligned}$$

Source voltage (phasor complex RMS value):

$$\underline{V}_s = 16 \cdot e^{j45^\circ} = (11.3137 + j11.3137) \text{ V}$$

Source current (phasor complex RMS value):

$$\underline{I}_s = \frac{\underline{V}_s}{\underline{Z}} = \frac{16}{8.1901} \cdot e^{j(45^\circ - 5.39^\circ)} = 1.9536 \cdot e^{j39.61^\circ} = (1.5051 + j1.2455) \text{ A}$$

a) Complex power delivered by the source:

$$\underline{S} = \underline{V}_s \underline{I}_s^* = (11.3137 + j11.3137) \text{ V} \cdot (1.5051 - j1.2455) \text{ A} = (31.12 + j2.94) \text{ VA}.$$

b) Active power delivered by source:

$$P = \text{Re}(\underline{S}) = 31.12 \text{ W}$$

c) Reactive power:

$$Q = \text{Im}(\underline{S}) = 2.94 \text{ VAR}$$

d) Apparent power:

$$S = \sqrt{P^2 + Q^2} = 31.26 \text{ VA}.$$

e) Power factor (scalar) and type:

$$\cos \varphi = \frac{P}{S} = \frac{31.12}{31.26} \approx 0.9956.$$

Since $Q > 0$ the net load is inductive, so the power factor is *lagging*.

Exercise 3: RMS Values and Average Power

The voltage across a load and the current through it are given by $v(t) = 20 + 60 \cos(100t)$ V and $i(t) = 1 - 0.5 \sin(100t)$ A. Find:

- the RMS values of the voltage and of the current
- the average power dissipated in the load

Solution

- a) For a signal composed of a constant (DC) term plus a single sinusoid,

$$x(t) = X_{\text{DC}} + \hat{x} \cos(\omega t + \phi),$$

the mean square (time average of $x^2(t)$) over one period is

$$\overline{x^2(t)} = X_{\text{DC}}^2 + \frac{\hat{x}^2}{2},$$

because the cross term between DC and the sinusoid averages to zero and the sinusoid contributes $\hat{x}^2/2$. Thus the RMS value is

$$X = \sqrt{X_{\text{DC}}^2 + \frac{\hat{x}^2}{2}}.$$

Apply this to the voltage:

$$V_{\text{DC}} = 20, \quad \hat{v} = 60$$

Hence

$$V = \sqrt{20^2 + \frac{60^2}{2}} = \sqrt{2200} = 10\sqrt{22} \approx 46.9 \text{ V}.$$

Apply the same formula to the current. Write the current as

$$i(t) = I_{\text{DC}} + \hat{i} \cos(100t + \phi_i).$$

Here $I_{\text{DC}} = 1$. The AC term is $-0.5 \sin(100t)$, which has amplitude $\hat{i} = 0.5$. (The sign and phase do not affect the RMS magnitude.) Thus

$$I = \sqrt{1^2 + \frac{(0.5)^2}{2}} = \sqrt{1.125} \approx 1.061 \text{ A}.$$

- b) The instantaneous power is

$$p(t) = v(t)i(t) = (20 + 60 \cos(100t))(1 - 0.5 \sin(100t)).$$

Expand:

$$\begin{aligned} p(t) &= 20 \cdot 1 + 20 \cdot (-0.5 \sin(100t)) + 60 \cos(100t) \cdot 1 + 60 \cos(100t) \cdot (-0.5 \sin(100t)) \\ &= 20 - 10 \sin(100t) + 60 \cos(100t) - 30 \cos(100t) \sin(100t) \end{aligned}$$

The average power P is the time average of $p(t)$ over one period. Every pure sinusoid (or product of sinusoids of the same frequency that reduces to a sinusoid at twice the

frequency) has zero average. Thus the time average of $-10 \sin(100t)$, of $60 \cos(100t)$, and of $-30 \cos(100t) \sin(100t)$ is zero. Therefore only the DC term remains:

$$P = \overline{p(t)} = 20 \text{ W}$$

Alternatively, using the decomposition into DC and sinusoidal parts:

$$P = V_{\text{DC}} I_{\text{DC}} + \frac{1}{2} \hat{v} \hat{i} \cos \varphi,$$

where φ is the phase difference between the voltage and current sinusoids. Here $V_{\text{DC}} I_{\text{DC}} = 20 \cdot 1$ and the sinusoidal part has

$$\hat{v} = 60, \quad \hat{i} = 0.5.$$

Voltage sinusoid: $60 \cos(100t)$ (phase 0). Current sinusoid: $-0.5 \sin(100t) = 0.5 \cos(100t + \frac{\pi}{2})$ (phase $+\frac{\pi}{2}$). Thus $\varphi = 0 - (+\frac{\pi}{2}) = -\frac{\pi}{2}$ and $\cos \varphi = \cos(-\frac{\pi}{2}) = 0$. Hence the sinusoidal contribution is zero and

$$P = 20 \text{ W}.$$

Exercise 4: Reactive Power Compensation

An AC motor with impedance $\underline{Z} = 4.2 + j 3.6 \Omega$ is supplied by a 220-V, 60-Hz source.

- Find the power factor, the real power P , and the reactive power Q .
- Determine the capacitor required to be connected in parallel with the motor so that the power factor is corrected to unity.

Solution

- The magnitude and angle of the load impedance are

$$|\underline{Z}| = \sqrt{R^2 + X^2} = \sqrt{4.2^2 + 3.6^2} \approx 5.532 \Omega,$$

$$\varphi = \arctan \frac{X}{R} = \arctan \frac{3.6}{4.2} \approx 0.709 \text{ rad} \approx 40.6^\circ.$$

The RMS current is

$$I = \frac{V}{|\underline{Z}|} = \frac{220}{5.532} \approx 39.77 \text{ A}.$$

The power factor (PF) is the cosine of the impedance angle (lagging because $X > 0$):

$$\cos \varphi \approx 0.76 \quad (\text{lagging}).$$

Compute the complex power:

$$\underline{S} = \underline{V} \underline{I}^* = \underline{V} \frac{\underline{V}^*}{\underline{Z}^*} = \frac{V^2}{\underline{Z}^*} = V^2 / (R - jX)^* = \frac{V^2(R + jX)}{R^2 + X^2}$$

So the real and reactive powers are

$$P = \frac{V^2 R}{R^2 + X^2} \approx 6.643 \text{ kW},$$

$$Q = \frac{V^2 X}{R^2 + X^2} \approx 5.694 \text{ kVAR}.$$

- b) To correct the power factor to unity we must supply capacitive reactive power Q_C that cancels the motor's inductive reactive power Q . For a capacitor connected in parallel across the same RMS voltage V ,

$$|Q_C| = V^2 \omega C, \quad \omega = 2\pi f.$$

We need $|Q_C| = Q$, hence

$$C = \frac{Q}{V^2 \omega} = \frac{5694}{220^2 \cdot 2\pi \cdot 60} \approx 312 \mu\text{F}.$$

Exercise 5: Series Resonance

A series RLC network has $R = 2 \text{ k}\Omega$, $L = 40 \text{ mH}$, and $C = 1 \mu\text{F}$. Calculate the impedance at resonance and at one-fourth, one-half, twice, and four times the resonant frequency.

Solution

For a series RLC network with

$$R = 2 \text{ k}\Omega, \quad L = 40 \text{ mH}, \quad C = 1 \mu\text{F}$$

the series impedance is

$$\underline{Z}(j\omega) = R + j(\omega L - \frac{1}{\omega C}).$$

Resonance occurs when the net reactance is zero:

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Substitute the values:

$$\omega_0 = \frac{1}{\sqrt{(40 \times 10^{-3})(1 \times 10^{-6})}} = 5000 \text{ rad s}^{-1}$$

For a given angular frequency ω ,

$$X_L = \omega L, \quad X_C = \frac{1}{\omega C}, \quad X = X_L - X_C,$$

$$\underline{Z} = R + jX, \quad |\underline{Z}| = \sqrt{R^2 + X^2}, \quad \varphi = \arctan \frac{X}{R}.$$

We evaluate at $\omega = k\omega_0$ for $k = \frac{1}{4}, \frac{1}{2}, 1, 2, 4$.

k	ω (rad/s)	X_L (Ω)	X_C (Ω)	$\underline{Z}$ (k Ω)	$ \underline{Z} $ (k Ω)	φ
0.25	1250	50	800	$2 - j0.75$	2.136	-20.6°
0.5	2500	100	400	$2 - j0.3$	2.022	-8.5°
1	5000	200	200	2	2	0°
2	10000	400	100	$2 + j0.3$	2.022	$+8.5^\circ$
4	20000	800	50	$2 + j0.75$	2.136	$+20.6^\circ$

Observations:

- At resonance ($k = 1$) the inductive and capacitive reactances cancel, leaving purely resistive impedance $Z = R = 2000 \Omega$.
- The network is capacitive ($X < 0$) for frequencies below resonance and inductive ($X > 0$) for frequencies above resonance.
- The magnitudes at pairs $k = \frac{1}{4}$ and $k = 4$ are equal (symmetry), and likewise for $k = \frac{1}{2}$ and $k = 2$.

Exercise 6: Parallel Resonance

Rework Exercise 6 if the elements are connected in parallel.

Solution

For the parallel connection the total admittance is

$$\underline{Y}(j\omega) = \underbrace{\frac{1}{R}}_G + j \left(\underbrace{\omega C}_{B_C} + \underbrace{\left(-\frac{1}{\omega L}\right)}_{B_L} \right) = G + jB(\omega).$$

The impedance is the reciprocal of the admittance:

$$\underline{Z}(\omega) = \frac{1}{\underline{Y}(\omega)} = \frac{1}{G + jB} = \frac{G - jB}{G^2 + B^2}$$

Its magnitude and phase are

$$|\underline{Z}| = \frac{1}{\sqrt{G^2 + B^2}}, \quad \varphi = \text{atan2}(\text{Im}(\underline{Z}), \text{Re}(\underline{Z})) = \arctan \frac{-B}{G}.$$

Parallel resonance (net susceptance zero) occurs at the same angular frequency as series resonance:

$$B(\omega_0) = 0 \quad \Rightarrow \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

With the given values

$$\omega_0 = \frac{1}{\sqrt{(40 \times 10^{-3})(1 \times 10^{-6})}} = 5000 \text{ rad s}^{-1}.$$

Also

$$G = \frac{1}{R} = \frac{1}{2000} = 5 \times 10^{-4} \text{ S}.$$

At resonance $B(\omega_0) = 0$, so

$$Y(\omega_0) = G \quad \Rightarrow \quad Z(\omega_0) = \frac{1}{G} = R = 2000 \Omega.$$

We compute for each ratio k :

$$\omega = k\omega_0, \quad B = \omega C - \frac{1}{\omega L}, \quad \underline{Z} = \frac{G - jB}{G^2 + B^2}$$

k	ω (rad/s)	B_C (m Ω)	B_L (m Ω)	$\underline{Z}$ (Ω)	$ \underline{Z} $ (Ω)	φ
0.25	1250	1.25	20	$1.42 + j 53.30$	53	$+88.5^\circ$
0.5	2500	2.5	10	$8.85 + j 132.74$	133	$+86.2^\circ$
1	5000	5	5	2000	2000	0°
2	10000	10	2.5	$8.85 - j 132.74$	133	-86.2°
4	20000	20	1.25	$1.42 - j 53.30$	53	-88.5°

Notes on the results:

- At resonance the susceptances of the inductor and capacitor cancel and the parallel branch behaves as a pure conductance G , giving $Z(\omega_0) = R$.
- For frequencies below resonance ($k < 1$) the net susceptance B is negative (inductive behavior), producing an impedance with a large positive imaginary part (inductive reactance).
- For frequencies above resonance ($k > 1$) B is positive (capacitive behavior), producing impedance with a large negative imaginary part (capacitive reactance).
- The impedances occur in conjugate pairs for reciprocal frequency ratios (e.g. $k = 0.25$ and $k = 4$; $k = 0.5$ and $k = 2$).